

Milestone 1: NY State Report Card

For this milestone we're exploring the New York State School Report Card 2024-25 release.

We want to use this dataset to look at three questions:

1. How does money affect pupil performance at these schools?
2. How have test scores trended over the last few years, and what does the increase/dropoff look like for affluent schools versus poorer schools? And how does it compare in math versus reading?
3. In what ways are disadvantaged schools disadvantaged versus their peers?

With more data, we'd also like to look at how AI and social media lead to performance changes of students over time.

Data loading

The database has around 30 tables. The ones that look useful for the questions are the math and ELA exam tables, the expenditures table, the chronic absenteeism table, and the two teacher tables (inexperienced teachers and out-of-certification teachers).

The exam tables are pretty big (around 650k rows each) because every row is a school $\times$ assessment $\times$ subgroup combination. The expenditure and teacher tables are at the district level, which is one row per district per year, so they're much smaller.

The dataset only has 2024 and 2025. So when we say trends, we really only have two data points to work with for this milestone. For the final, we want to grab earlier SRC releases.

The exam tables already break each result down into a long list of subgroups (Economically Disadvantaged, English Language Learner, race/ethnicity, gender, homeless, foster care, etc.). That's helpful, because it already provides a definition of disadvantaged.

MATH3_8 is a roll-up summary, but the MEAN_SCORE column is empty for those rows. The grade-level rows (MATH3 through MATH8) are the ones that actually have scale scores. So we'll filter to those when we want a number to compare.

ENTITY_CD column is a 12-digit code that identifies both districts and schools. District codes end in 0000, while school codes don't. The expenditures table mixes both kinds of rows. we'll filter to districts only when joining because we only care about the aggregate

Around 700 districts and 4500 schools.

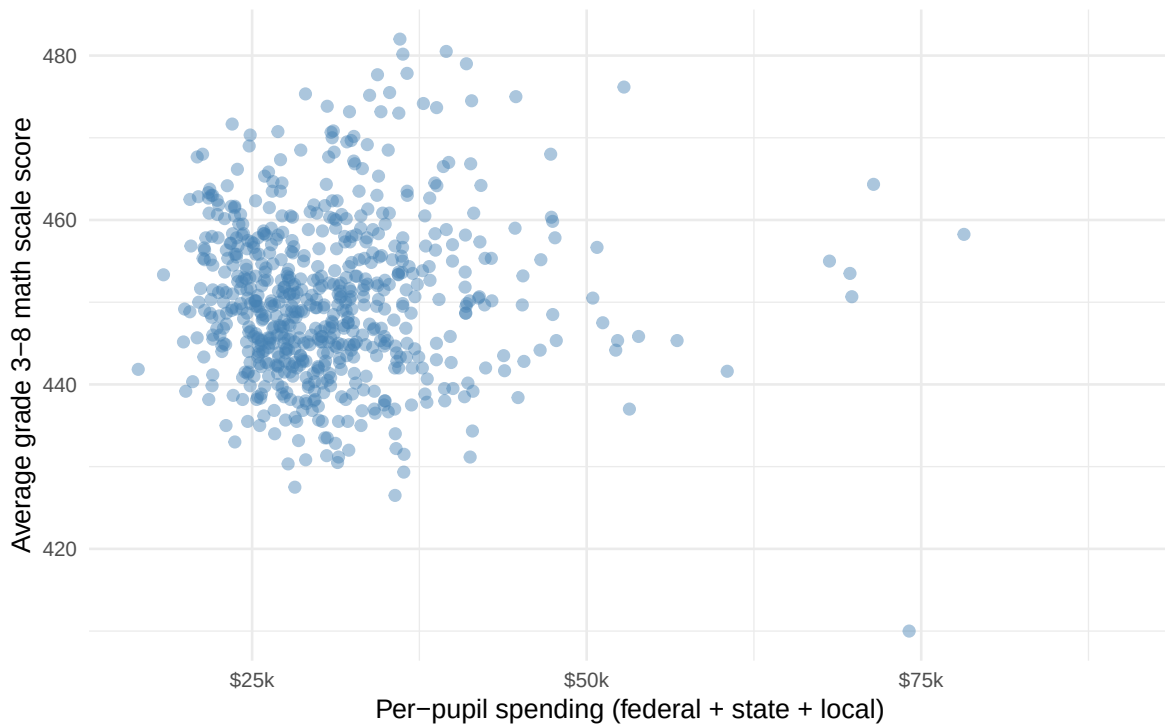
Goal 1: Does money buy performance?

Our first instinct was a simple scatter: do districts that spend more per pupil score higher?

The puzzle: spending and scores move in the wrong direction

Spending doesn't predict scores. If anything, the pattern runs backwards

Each point is a NY school district, 2024–25.

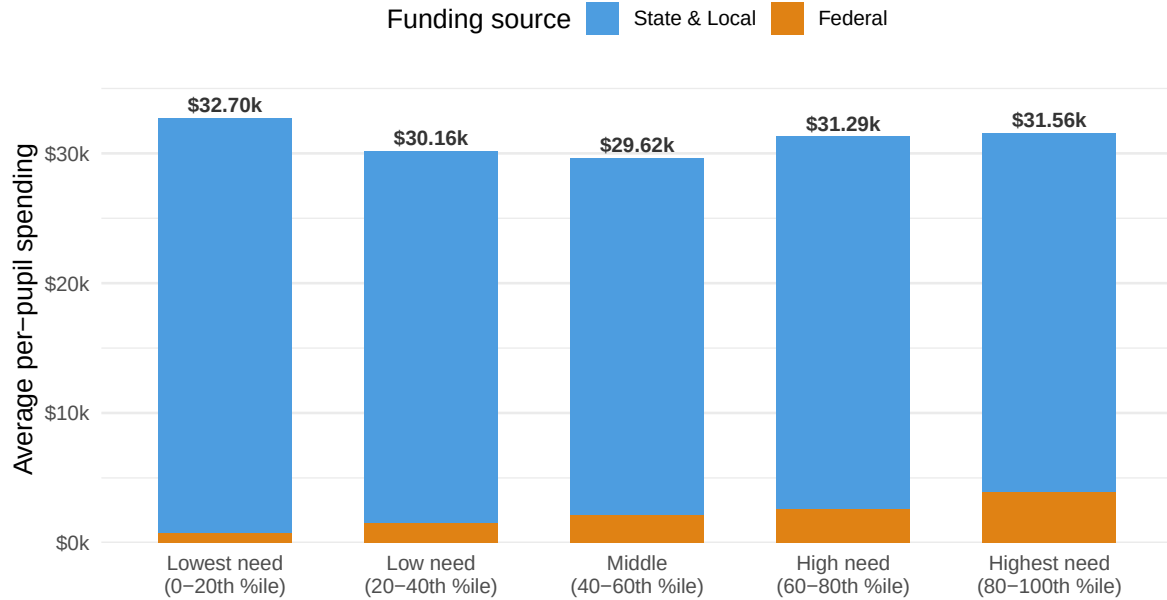


There's no upward pattern here. If anything, higher spending is weakly associated with *lower* scores, the opposite of what we expected. So what's going on?

Why high-need districts spend more: equity funding at work

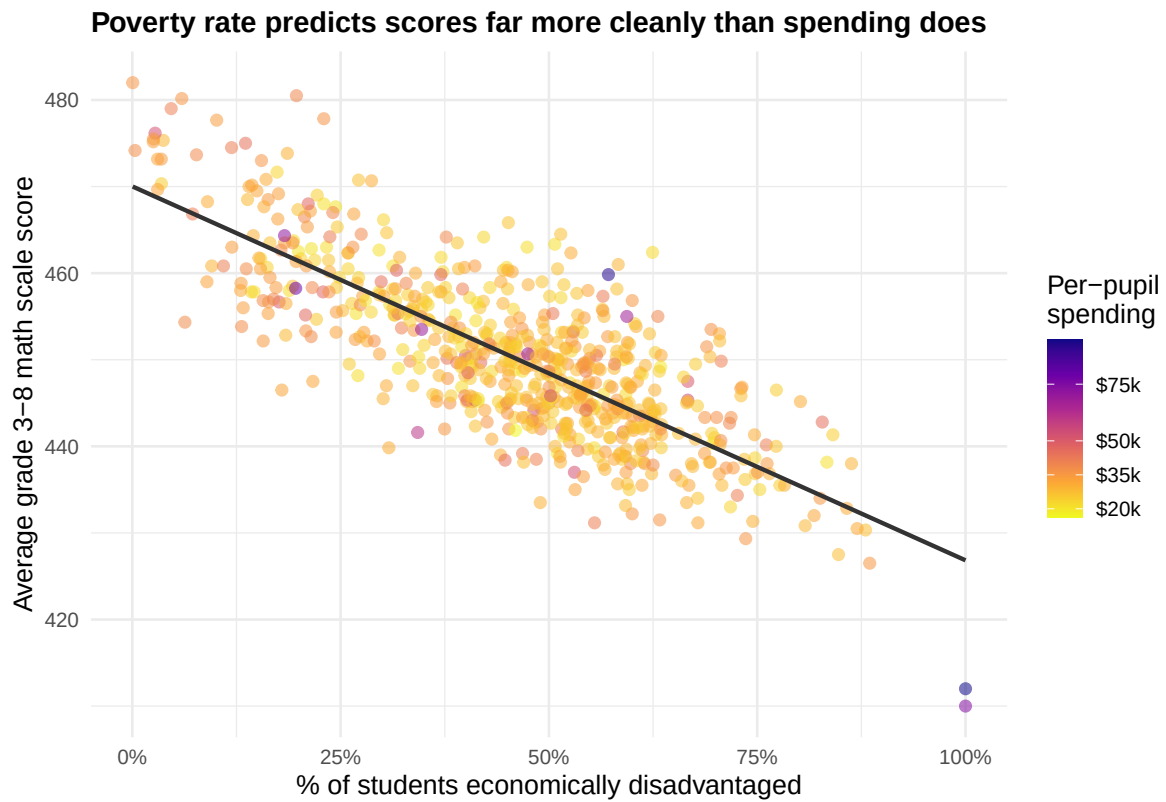
Equity aid flows to high-need districts, but state & local funding barely changes

Average per-pupil spending by district needs quintile, 2024–25.



High-need districts do receive more money overall, mainly through federal aid. But that extra funding hasn't translated into higher scores, which suggests money alone isn't enough to close the gap.

The real driver: poverty, not spending



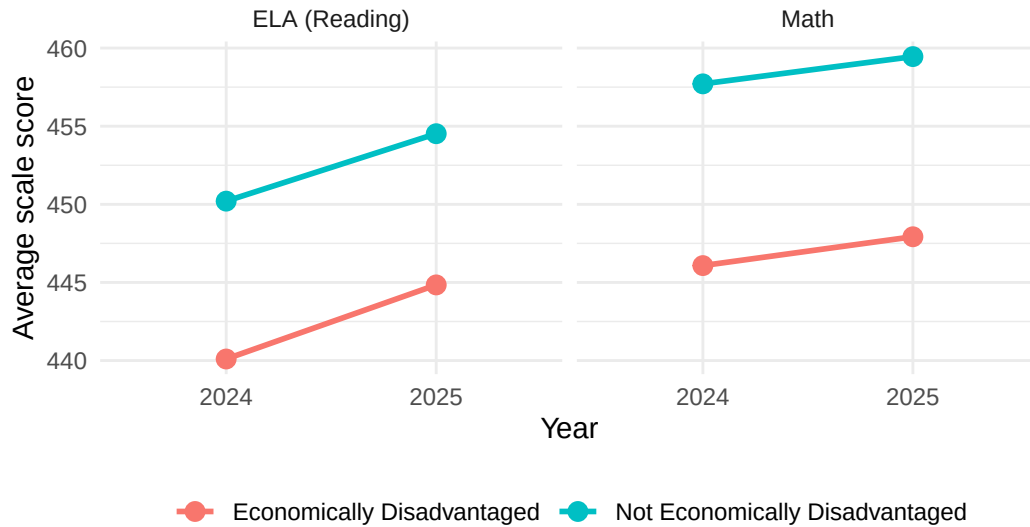
Poverty rate is a strong predictor of scores ($r = -0.79$), far more so than spending ($r = -0.02$). That raises the next question: how wide is this gap, and is it closing?

Goal 2: Affluent vs poorer schools in math and reading

If poverty predicts scores more cleanly than spending, the next question is how wide that gap actually is, and whether it's moving.

A 13-point gap that doesn't budge

The gap between disadvantaged and not is wide and not closing
Average grade 3–8 scale score, NY State



Disadvantaged students sit roughly 13–14 scale-score points behind their peers in both subjects, and the gap barely moves between 2024 and 2025. Math runs a hair wider than ELA, but the two subjects don't share a scale, so that's a soft observation.

The longer view: COVID broke the trend

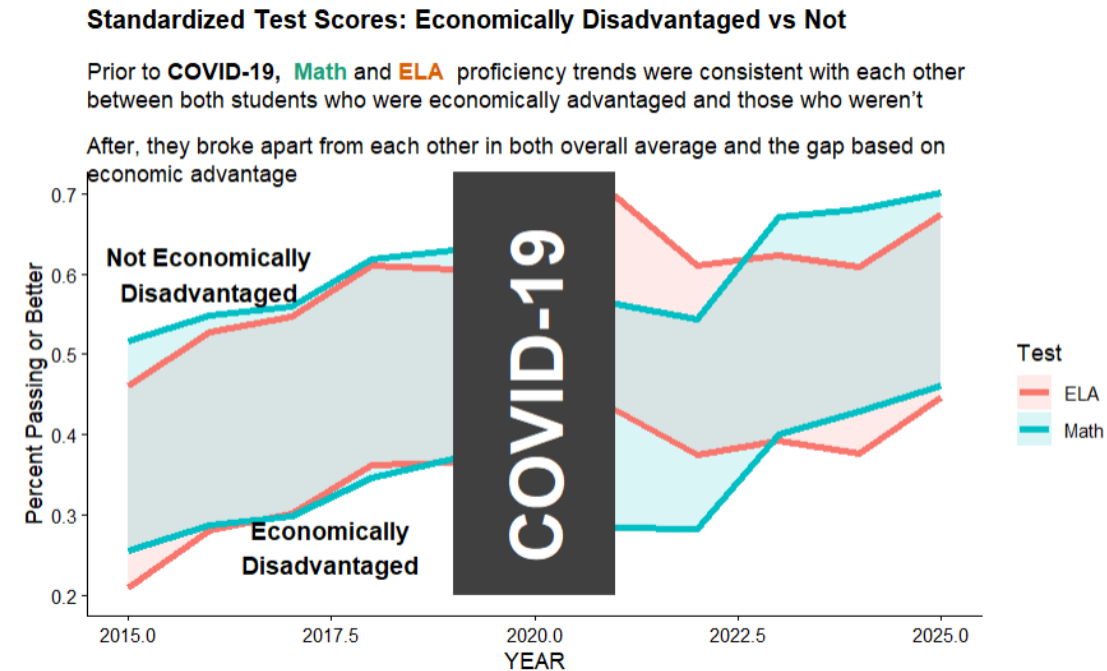


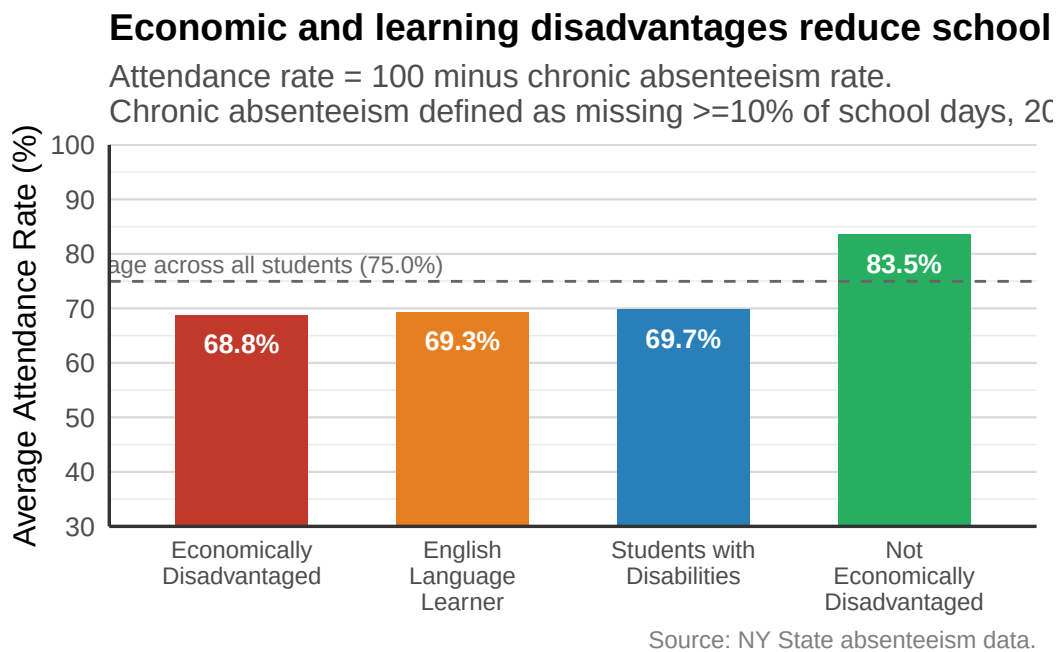
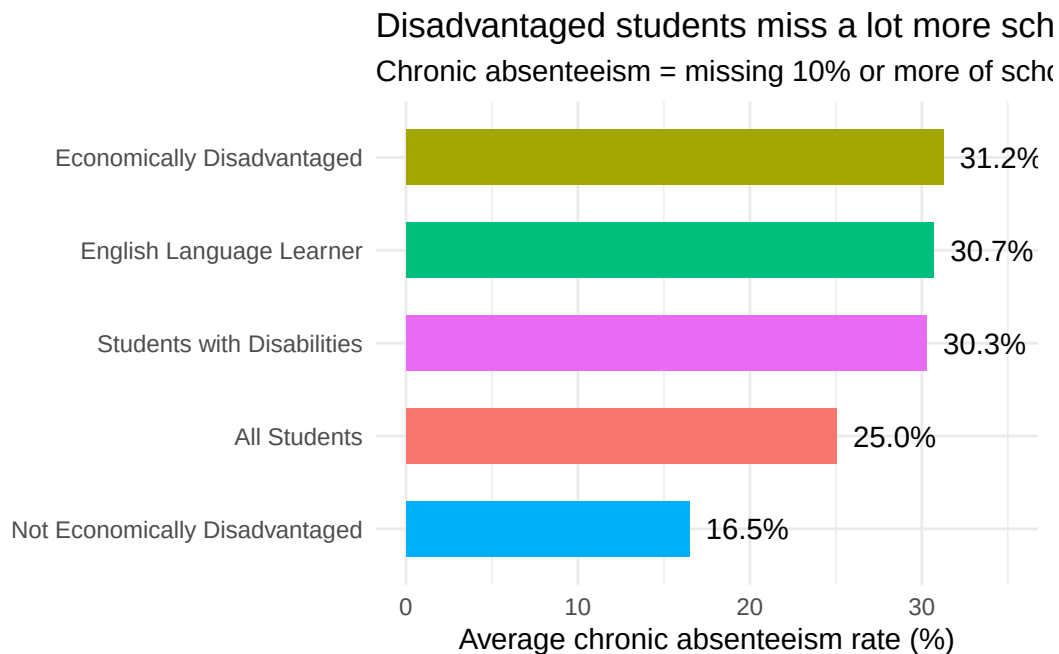
Figure 1: Standardized test scores: economically disadvantaged vs not, ELA and Math, 2015–2025

Before the pandemic, both groups climbed together. After, math fell harder than ELA for economically disadvantaged students, and they haven't recovered to their pre-pandemic trajectory the way their peers have. The flat 2024–2025 gap isn't steady state. It's the tail end of a bigger setback.

Goal 3: How are disadvantaged schools disadvantaged?

A gap this wide and this stuck rarely runs on student effort alone. Two structural mechanisms are measurable in this dataset: how often students show up, and who's teaching them.

Showing up isn't equal either



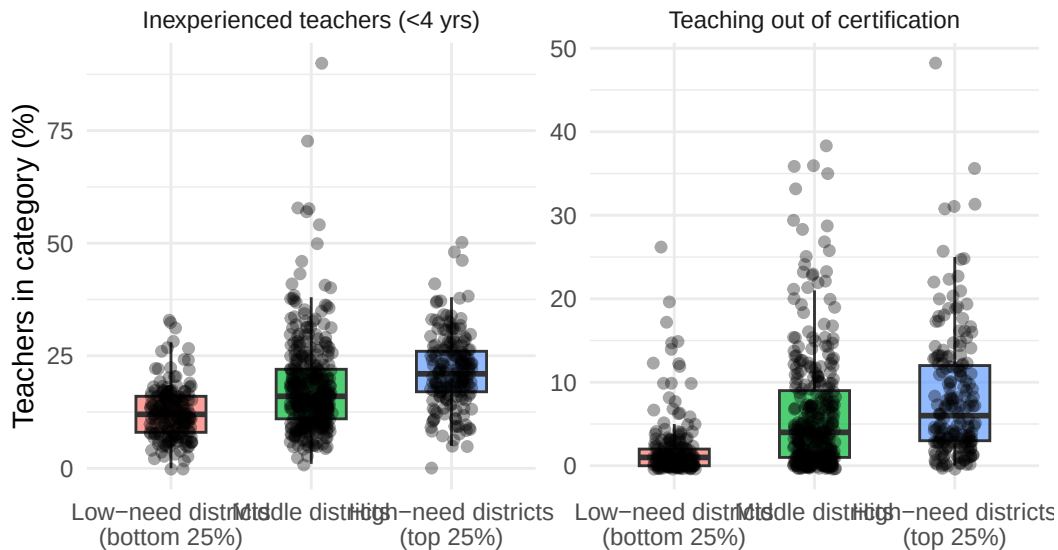
Economically disadvantaged students miss roughly twice as much school as their peers. You can't close an academic gap from outside the building, which makes absenteeism a plausible

cause, not just a symptom.

High-need districts shoulder rougher staffing too

Teacher data doesn't break down by student subgroup, so we shift to the district level: do high-need districts get a worse staffing deal?

High-need districts tend to have more teacher staffing instability Districts are grouped by their share of economically disadvantaged student



Each jittered point is one district and the boxplots summarize the distribution within each need group. The pattern isn't identical district to district, but both metrics shift upward as need rises, with a noticeably heavier right tail. Disadvantaged districts face structural challenges that go beyond student test scores, including less stable teacher staffing. Combined with absenteeism, the score gap isn't only about what students bring to school; it's about what school brings to them.